Opening lightning talk

When does structure make reasoning trustworthy?

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1 Reasoning in natural language

2 Symbolic translation

3 Results

4 Other applications

Reasoning with natural language

Story

A is at 3 o'clock from N. B is above A. S is south of B. G is east of S.

Where is G relative to N?

→

LLM REASONS IN TEXT

"…G falls between 12 and 9 o'clock, so G is slightly below N"

→

✗ **G is below N**

hallucinated relation (answer: right of N)

LESS ACCURATE AS COMPLEXITY GROWS

93% → 27%

accuracy from 1 to 10 hops (LLaMA-3.1-70B, StepGame)

NOT TRUSTWORTHY: WHAT WE CHECK

Hallucinated relations	**17%** of text failures
Grounded in evidence (faithfulness)	**52%** of answers
Logically consistent (plausibility)	**42%** of answers

Paper *Rajpal, Premsri & Kordjamshidi. Spatial Reasoning via Modality Switching Between Language and Symbolic Representations. Findings of EMNLP 2026.*

Translating text to symbolic representations

Spatial story

A is at 3 o'clock from N. B is above A. S is south of B. G is east of S.

Question

Where is G relative to N?

→

Relational triples

LLM extraction, sentence by sentence

head	relation	tail
A	right	N
B	above	A
S	below	B
G	right	S

→

Full grid

	col 1	col 2	col 3
row 1	–	B	–
row 2	**N**	A/S	**G**

whole scene

Coordinates

```
N (0,0)   A (1,0)
B (1,1)   G (2,0)
```

Prolog constraint solver

Pruned grid

	col 1	col 2
row 1	–	–
row 2	**N**	**G**

question entities only

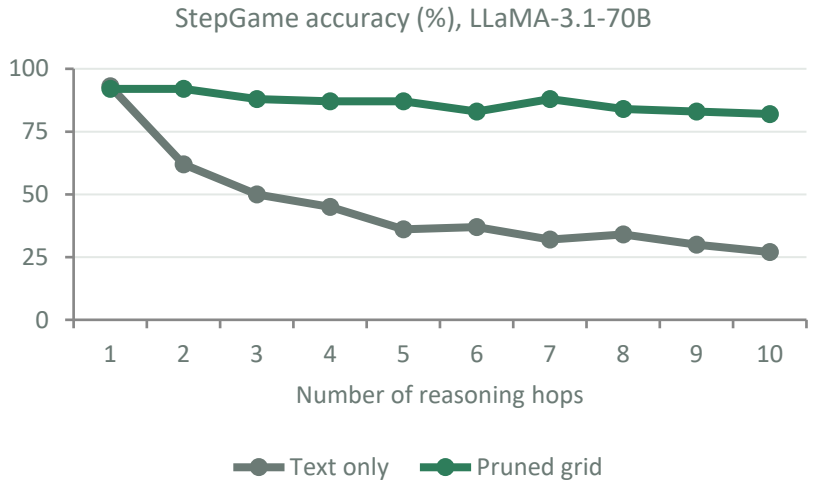
Reasoning with triples

Grids and coordinates are built from the triples by code

Paper *Rajpal, Premsri & Kordjamshidi. Spatial Reasoning via Modality Switching Between Language and Symbolic Representations. Findings of EMNLP 2026.*

With symbolic representations

STABLE ACROSS COMPLEXITY



MORE ACCURATE, FEWER ERRORS

+42% accuracy

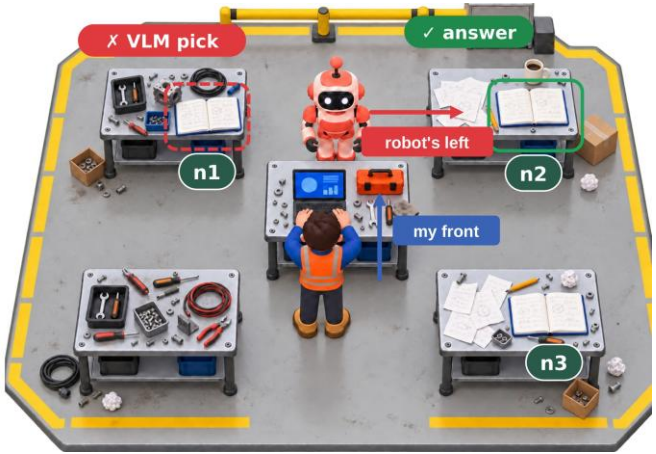
64% hallucinations fixed

82% Evidence-grounding errors fixed

→ More trustworthy and more accurate reasoning

Paper *Rajpal, Premsri & Kordjamshidi. Spatial Reasoning via Modality Switching Between Language and Symbolic Representations. Findings of EMNLP 2026.*

Translating images to symbolic representations

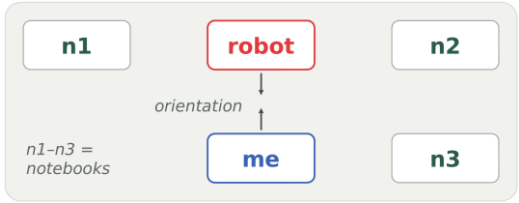


*"Bring me the notebook **on your left** and **in front of me**"*

Reasoning in code

1 **3D scene parsing**

3D Positions + Orientations



2 **Soft symbolic scores**

each relation in its own frame

object	notebook	robot's left	my front
n2	**0.98**	**0.94**	**0.89**
n3	0.96	0.91	0.04

3 **Symbolic program**

written by an LLM, then executed

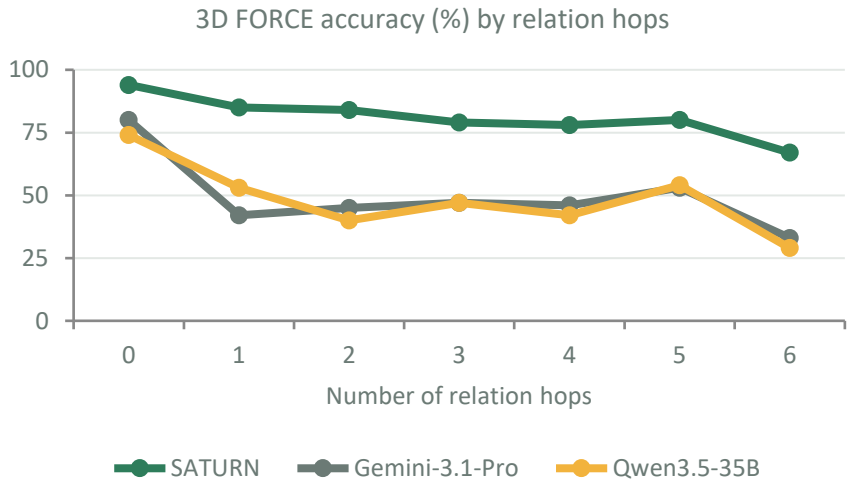
```
is_notebook(x)
& obj_left(x, robot)
& obj_front(x, me)
```

✓ n2

Paper *Kamali, Premsri, **Rajpal**, Zadeh, Li & Kordjamshidi. SATURN: Symbolic Spatial Reasoning for Multi-Perspective Grounding. EMNLP 2026.*

With symbolic representations of images

STABLE ACROSS INCREASING COMPLEXITY



MORE ACCURATE

+25.1% vs. 57.8% for the best VLM on 3D FORCE

→ Training-free, more accurate and more stable than end-to-end VLMs

Paper *Kamali, Premsri, **Rajpal**, Zadeh, Li & Kordjamshidi. SATURN: Symbolic Spatial Reasoning for Multi-Perspective Grounding. EMNLP 2026.*

Beyond spatial reasoning



Spatial

Stories, scenes

↓

Grids, 3D scenes, programs



Legal

Contract clauses

↓

Rule graphs with conditions & exceptions



Temporal

Event descriptions

↓

Event logic trees over time



Structured data

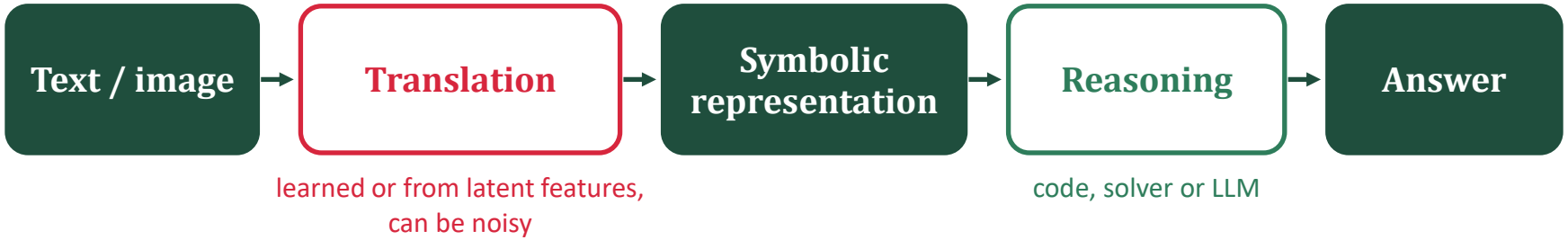
Questions

↓

SQL, knowledge graphs

Same recipe: translate → structure → execute. **Same risk:** a wrong structure still executes.

Open question: the translation step



Our grids: 100 F1 from gold relations, 87–90 F1 from extracted relations (StepGame)

How do we improve the translation?

How do we build metrics to test it without gold structures?